

Battery Heating

Low temperature will not be a barrier to battery operation any more



Storage&Battery

Introduction

In some high-latitude or high-altitude areas, the temperature (during winter) is quite low. The battery charging and discharging performance will be impacted significantly by the low temperature. SolaX developed a function of “Battery Heating” to make your ESS able to work under extreme low temperature.

This function is disabled by default, so you need to enable this function if needed.

After you enable this function, you need to set heating periods

In heating period

The cell temperature is among $-30 \sim 0^{\circ}\text{C}$, it will be heated until 10°C . Then, heating will stop, once the cell temperature is below 0°C again, the battery system will be heated again.

The heating system will take energy from PV in priority. If the PV power is not enough, it will take energy from battery or grid, which depends on the **BAT actual SOC**.

If **BAT actual SOC** $> 35\%$, it takes from **Battery**.

If **BAT actual SOC** $< 35\%$, it takes from **Grid**.

Out of heating period

The battery will be heated only if the PV has surplus energy.

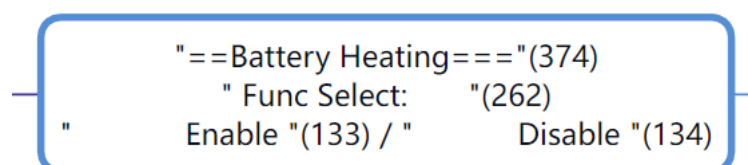
Note:

- Only if the cell temperature is higher than -10°C , the battery will be able to discharge, and only if cell temperature is higher than 0°C , the battery will be able to be charged.*
- The heating power per module is 180W for T30 battery, and 120W for HS25/HS36 battery.*
- Please check the heating wires are correctly connected, or the heating function will not take effect.*
- The heating control is based on the Cell temperature, which the internal temperature rather than the environment temperature. Normally the cell temperature is higher than the environment.*

Configuration

Setting path:

Main → Settings → Advance Setting → Battery Heating → Func Select



Enable it and set heating periods.

Heating Period 1 (375)
Start Time (343)
00:00

Heating Period 1
End Time
00:00

Heating Period 2
Start Time
00:00

Heating Period 2
End Time
00:00

Compatibility

Hardware version:

X1-Hybrid/Fit G4

T-BAT-SYS-HV 3.0

X3-Hybrid/Fit G4

T-BAT-SYS-HV S2.5/S3.6

Inverter software version:

ARM $\geq$ 1.19

DSP $\geq$ 1.22

Battery software version:

3.05 for T30 master (MC0600)